

ARATA HORIE

507 (5F), CCR Bldg., 4-6-1 KOMABA MEGURO-KU, TOKYO 153-8505, JAPAN

◇ a.horie@commissure.co.jp

EDUCATION

The University of Tokyo

Ph.D. in Engeneering

Department of Advanced Interdisciplinary Studies

Advisor: Dr. Masahiko Inami

Tokyo, Japan

April 2019 - March 2022

Tohoku University

M.S. in Information Science

Department of Applied information science

Advisor: Dr. Masashi Konyo

Sendai, Japan

April 2017 - March 2019

Tokyo University of Science

B.S. in Mechanical Engineering

Department of Mechanical Engineering

Advisor: Dr. Hiroshi Mizoguchi

Tokyo, Japan

April 2013 - March 2017

WORK EXPERIENCE

commissure Inc.

CTO

Tokyo, Japan

January 2023 - present

Keio University Graduate School of Media Design, Embodied Media

Project Senior Assistant Professor

Tokyo, Japan

April 2026 - present

Keio University Graduate School of Media Design, Embodied Media

Project Assistant Professor

Tokyo, Japan

April 2023 - March 2026

The University of Tokyo, Research Center for Advanced Science and Technology, Information Somatics Lab

Project Assistant Professor

Tokyo, Japan

April 2022 - March 2023

Information Somatics Lab, The University of Tokyo

Research Assistant / JSPS Research Fellow DC2, Advisor: Masahiko Inami

Tokyo, Japan

April 2019 - March 2022

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Haptic Research Internship

Tokyo, Japan

April 2019 - September 2019

Human-Robot Informatics Lab, Tohoku University

Research Assistant, Advisor: Masashi Konyo

Sendai, Japan

April 2017 - March 2019

GRANTS

ACT-X Hardware in Future for Resilience of Real Space, "Design of Self-other Inseparable Intercorporeality Through Body Surface Deformation Device", JST

2021 - 2023

Japan Science and Technologies

Japan Society for Promotion of Science DC2, "Self-motion Sensation Induction by the Presentation of Distributed Force", JSPS

2020 - 2021

Japan Society for Promotion of Science

HONORS AND AWARDS

- Best Paper Award, CHI 2025** **2025**
Co-author, "It's Like Being On Stage" : Conveying Dancers' Expressiveness Through A Haptic-Installed Contemporary Dance Performance"
- Best Demo Award, SIGGRAPH Asia 2025 Emerging Technologies** **2025**
Co-author, "Experience Sharing Box: A Platform for Capturing, Storing, and Playback of Multisensory Memory"
- Official Selection for Laval Virtual, SIGGRAPH 2024 Emerging Technologies** **2024**
Co-author, "FEELTECH Wear: Enhancing Mixed Reality Experience with Wrist to Finger Haptic Attribution"
- Best Demo Award, EuroHaptics Conference 2024** **2024**
Co-author, "Rotational Skin-stretch Distribution Creates Directional Force Sensation on the Wrist"
- Dean's Award, Graduate School of Engineering** **2022**
The University of Tokyo
- Fellowships from Japan Society for Promotion of Science DC2, JSPS** **2020 - 2021**
Japan Society for Promotion of Science
- Best Paper Award, UbiComp MIMSVAI 2021** **2021**
Co-author, "High-speed non-contact thermal display using infrared rays and shutter mechanism"
- Best Paper Award Nominated, World Haptics Conference 2021** **2021**
Author, "Two-dimensional moving phantom sensation created by rotational skin stretch distribution"
- Outstanding Performance Award, Asia Digital Art Award 2020** **2020**
Author, "TorsionCrowds"
- Disruptive Challenge Finalist, INNO-Vation 2019** **2019**
Author, Ministry of Internal Affairs and Communications in Japan
- Best Demonstration Award Gold Winner, AsiaHptics 2018** **2018**
Author, "Enhancing Haptic Experience in a Seat with Two-DoF Buttock Skin Stretch"
- Ultrahaptics Student Challenge Finalist, EuroHaptics 2018** **2018**
Co-author, "Weather Teleport"
- Robomech Award, Robotics and Mechatronics 2018** **2018**
Author, "Induction of acceleration sensation of self-motion by presenting shear force to buttocks"
- Young Researcher's Award, VRSJ 2018** **2018**
Author, "Presenting a pushing up feeling from the seat by buttocks skin stretch"
- Excellent Presentation Award, SICE SI 2018** **2018**
Author, "2-DoF Buttock Skin Stretch for Inducing Acceleration Perception of Self-motion"
- COLOPL Award, International collegiate Virtual Reality Contest 2017** **2017**
Team Manager, "Shall we coffee cup?"
- Young Researcher's Award, SICE SI 2017** **2017**
Author, "Sliding direction and speed presentation Combining high frequency and asymmetric vibration"
- Excellent Presentation Award, SICE SI 2017** **2017**
Author, "Sliding direction and speed presentation Combining high frequency and asymmetric vibration"

JOURNAL PAPER

1. Arata Horie, Hideki Shimobayashi, Hiroto Saito, and Masahiko Inami. Designing distributed-type haptic device based on rotational skin stretch. *Transactions of the Virtual Reality Society of Japan*, 25(4):402–411, 2020
2. Hiroto Saito, Arata Horie, Azumi Maekawa, Seito Matsubara, Sohei Wakisaka, Zendai Kashino, Shunichi Kasahara, and Masahiko Inami. Transparency in human-machine mutual action. *Journal of Robotics and Mechatronics*, 33(5):987–1003, 2021(Co-first author)

CONFERENCE PAPER

1. Emiri Yorioka, Yufan Zhu, Jinsuke Morita, Sohei Wakisaka, Arata Horie, and Kouta Minamizawa. Piano duo: Haptic sharing system supports two-piano practice. In *2025 IEEE World Haptics Conference (WHC)*, page TBD. IEEE, 2025
2. Ximing Shen, Xuan Li, Youichi Kamiyama, Danny Hynds, Giulia Barbareschi, RAY Lc, Sohei Wakisaka, Arata Horie, and Kouta Minamizawa. "it's like being on stage": Conveying dancers' expressiveness through a haptic-installed contemporary dance performance. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems*, pages 1–18, 2025(**Best Paper Award**)
3. Rodan Umehara, Arata Horie, and Kouta Minamizawa. Rotational skin-stretch distribution creates directional force sensation on the wrist. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 238–250. Springer, 2024
4. Arata Horie, Yunao Zheng, and Masahiko Inami. A wearable system integrating force myography and skin stretch feedback toward force skill learning. In *2023 IEEE World Haptics Conference (WHC)*, pages 190–196. IEEE, 2023
5. Ryo Murata, Arata Horie, and Masahiko Inami. Dynamic derm: Body surface deformation display for real-world embodied interactions. In *Proceedings of the Augmented Humans International Conference 2023*, AHs '23, page 267 – 277, New York, NY, USA, 2023. Association for Computing Machinery
6. Sosuke Ichihashi, Arata Horie, Masaharu Hirose, Zendai Kashino, Shigeo Yoshida, and Masahiko Inami. High-speed non-contact thermal display using infrared rays and shutter mechanism. *UbiComp '21*, page 565 – 569, New York, NY, USA, 2021. Association for Computing Machinery(**Best Paper Award**)
7. Arata Horie, Zendai Kashino, Hideki Shimobayashi, and Masahiko Inami. Two-dimensional moving phantom sensation created by rotational skin stretch distribution. In *2021 IEEE World Haptics Conference (WHC)*, pages 139–144, 2021(**Nominated to Best Paper Award**)
8. Arata Horie, MHD Yamen Saraiji, Zendai Kashino, and Masahiko Inami. Encounteredlimbs: A room-scale encountered-type haptic presentation using wearable robotic arms. In *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*, pages 260–269. IEEE, 2021
9. Hideki Shimobayashi, Tomoya Sasaki, Arata Horie, Riku Arakawa, Zendai Kashino, and Masahiko Inami. Independent control of supernumerary appendages exploiting upper limb redundancy. In *Augmented Humans Conference 2021*, AHs'21, page 19 – 30, New York, NY, USA, 2021. Association for Computing Machinery
10. Yudai Tanaka, Arata Horie, and Xiang 'Anthony' Chen. Dualvib: Simulating haptic sensation of dynamic mass by combining pseudo-force and texture feedback. In *26th ACM Symposium*

on *Virtual Reality Software and Technology*, VRST '20, New York, NY, USA, 2020. Association for Computing Machinery

11. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Buttock skin stretch: Inducing shear force perception and acceleration illusion on self-motion perception. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 135–147. Springer, 2018

DEMO

1. Haruomi Higashi, Yufan Zhu, Jack Brophy, Ismael Rasa, Jinsuke Morita, Arata Horie, Takatoshi Yoshida, and Kouta Minamizawa. Experience sharing box: A platform for capturing, storing, and playback of multisensory memory. In *Proceedings of the SIGGRAPH Asia 2025 Emerging Technologies*, pages 1–2. 2025 **SIGGRAPH Asia 2025 Emerging Technologies Best Demo Award**
2. Rodan Umehara, Harunobu Taguchi, Arata Horie, Yusuke Kamiyama, Shin Sakamoto, Hironori Ishikawa, and Kouta Minamizawa. Feeltech wear: Enhancing mixed reality experience with wrist to finger haptic attribution. In *ACM SIGGRAPH 2024 Emerging Technologies*, pages 1–2. 2024 **SIGGRAPH 2024 Emerging Technologies Official Selection for Laval Virtual**
3. Kiryu Tsujita, Takatoshi Yoshida, Kohei Kobayashi, Arata Horie, Nobuhisa Hanamitsu, and Kouta Minamizawa. Haptoroom: Using vibrotactile floor interfaces to enable reconfigurable haptic interaction onto any furniture surfaces. In *ACM SIGGRAPH 2024 Emerging Technologies*, pages 1–2. 2024
4. Rodan Umehara, Arata Horie, and Kouta Minamizawa. Rotational skin-stretch distribution creates directional force sensation on the wrist. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 238–250. Springer, 2024 **Best Demo Award**
5. Harunobu Taguchi, Youichi Kamiyama, Kenta Kan, Yulan Ju, Arata Horie, Yoshihiro Tanaka, Hironori Ishikawa, and Kouta Minamizawa. Multichannel haptic communication platform with wearable sensing and display. In *SIGGRAPH Asia 2023 Emerging Technologies*, pages 1–2. 2023
6. Aoi Uyama, Youichi Kamiyama, Sohei Wakisaka, Arata Horie, Tatsuya Saito, and Kouta Minamizawa. Somatic music: Enhancing musical experiences through the performer’s embodiment. In *SIGGRAPH Asia 2023 Posters*, pages 1–2. 2023
7. Arata Horie, Ryo Murata, Zendai Kashino, and Masahiko Inami. Seeing is feeling: A novel haptic display for wearer-observer mutual haptic understanding. In *ACM SIGGRAPH Asia 2022 Emerging Technologies*, SIGGRAPH Asia '22, New York, NY, USA, 2022. Association for Computing Machinery
8. Sosuke Ichihashi, Arata Horie, Masaharu Hirose, Zendai Kashino, Shigeo Yoshida, Sohei Wakisaka, and Masahiko Inami. Thermoblinds: Non-contact, highly responsive thermal feedback for thermal interaction. In *ACM SIGGRAPH 2022 Emerging Technologies*, SIGGRAPH '22, New York, NY, USA, 2022. Association for Computing Machinery
9. Arata Horie, Hideki Shimobayashi, and Masahiko Inami. Torsioncrowds: Multi-points twist stimulation display for large part of the body. In *ACM SIGGRAPH 2020 Emerging Technologies*, SIGGRAPH '20, New York, NY, USA, 2020. Association for Computing Machinery
10. Arata Horie, Masashi Konyo, Hikaru Nagano, and Satoshi Tadokoro. Whole-body perception of ground geometry by buttock skin stretch. In *IEEE World Haptics Conference*. IEEE, 2019

11. Arata Horie, Akito Nomura, Kenjiro Tadakuma, Masashi Konyo, Hikaru Nagano, and Satoshi Tadokoro. Enhancing haptic experience in a seat with two-dof buttock skin stretch. In *International AsiaHaptics conference*, pages 134–138. Springer, 2018 **Best Demo Award Gold Winner**
12. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Buttock skin stretch: Inducing shear force perception and acceleration illusion on self-motion perception. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*, pages 135–147. Springer, 2018

NON-REVIEWED PUBLICATIONS (FIRST AUTHOR)

1. Arata Horie and Masahiko Inami. Basic study of rotating tactors array presenting strain distribution on back skin. In *Annual Conference of the Virtual Reality Society of Japan*. VRSJ, 2019
2. Arata Horie, Akito Nomura, Kenjiro Tadakuma, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. 2-dof buttock skin stretch for inducing acceleration perception of self-motion. In *SICE System Integration*, volume 2018. SICE, 2018 **Excellent Presentation Award**
3. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Presenting pushing up feeling from the seat by buttocks skin stretch. In *Annual Conference of the Virtual Reality Society of Japan*, volume 2017. VRSJ, 2017 **Young Researcher's Award**
4. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Induction of acceleration sensation of self-motion by presenting shear force to buttocks. In *Robotics Mechatronics Conference 2018*. JSME, 2018 **Robomech Award**
5. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. -2nd report- sliding direction and speed presentation combining high frequency and asymmetric vibration. In *SICE System Integration*, volume 2017. SICE, 2017 **Excellent Presentation Award, Young Researcher's Award**
6. Arata Horie, Hikaru Nagano, Masashi Konyo, and Satoshi Tadokoro. Sliding direction and speed presentation combining high frequency and asymmetric vibration. In *Annual Conference of the Virtual Reality Society of Japan*, volume 2017. VRSJ, 2017

PATENT

力覚提示装置及び力覚提示方法 堀江新 (75/100) , 稲見昌彦 (25/100)	特願 2019-163973
力覚提示装置及び力覚提示方法 堀江新 (75/100) , 稲見昌彦 (25/100)	PCT/JP2020/032185
運動教示装置、運動教示システム、運動教示方法およびプログラム 堀江新 (45/100) , 檜山敦 (35/100) , 稲見昌彦 (20/100)	特願 2020-219284

TEACHING EXPERIENCE

Otherness and Interaction Technology in Embodied Interaction Keio University Graduate School of Media Design Guest Lecturer	November 2023 - Present
Digital Interaction Prototyping Keio University Graduate School of Media Design Lecturer	May 2023 - Present

Data Processing
Tohoku University
Teaching Assistant

October 2017 - February 2018